When Do AI Investments Create Competitive Advantage?

A Systematic Review of Firm-Level Empirical Evidence (2015–2026)

Bachelor's Thesis Proposal
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1 Introduction and Relevance

Artificial intelligence (AI), understood as machines that perform "cognitive functions that are usually associated with human minds, such as learning, interacting, and problem solving" (Raisch & Krakowski, 2021, p. 192)[1], has moved from research laboratories to the core of corporate investment agendas. Firms across industries are committing substantial resources to AI. The returns on these investments, however, remain uncertain. Nine of ten executives see a major business opportunity in AI, yet "most companies have a hard time generating value with AI" (Ransbotham et al., 2019, p. 1, as cited in Kemp, 2024, p. 619)[2]. Stock markets have on average even reacted negatively to firms' AI investments (Lui, Lee, & Ngai, 2022, as cited in Kemp, 2024, p. 619)[3].

Theory can make sense of this disconnect. AI "is regarded as a general-purpose technology akin to electricity, the steam engine, or the internet" (Kemp, 2024, p. 620)[4], and from the resource-based view a technology that every competitor can buy should not, by itself, be a source of sustained competitive advantage (Barney, 1991, p. 110)[5]. The advantage goes to the firm that combines the technology with resources others cannot

[1] [Raisch & Krakowski (2021), p. 192] "Artificial intelligence (AI) refers to machines performing cognitive functions that are usually associated with human minds, such as learning, interacting, and problem solving."

[2] [Kemp (2024), p. 619] "In a recent survey by Ransbotham, Khodabandeh, Fehling, LaFountain, & Kiron (2019: 1), nine out of 10 top managers reported that AI represents a large business opportunity for their firms and 43% of executives reported having implemented AI in their organizations. The report also noted, however, that 'most companies have a hard time generating value with AI."'

[3] [Kemp (2024), p. 619] "Lui, Lee, and Ngai (2022) offered empirical support for this concern, finding that markets penalize AI adoption at the firm level."

[4] [Kemp (2024), p. 620] "AI is regarded as a general-purpose technology akin to electricity, the steam engine, or the internet (Brynjolfsson & Mitchell, 2017; Frey & Osborne, 2013; Lynch, 2017)."

[5] [Barney (1991), p. 110] "If one firm can purchase these physical tools of production and thereby implement some strategies, then other firms should also be able to purchase these physical tools, and thus such tools should not be a source of sustained competitive advantage."

copy, such as its social relations and culture (Barney, 1991, p. 110)[6]. This thesis asks under what conditions firms achieve this combination, a question that matters to strategic management theory and to practitioners deciding which AI budgets create value.

2 Problem Definition and Research Gap

Empirical research linking firm-level AI investments to performance outcomes has grown quickly, but it is scattered across several literatures and points in different directions. Babina, Fedyk, He, and Hodson (2024, p. 1)[7] find higher growth and more product innovation among AI-investing firms, concentrated among firms that were already large (Babina et al., 2024, p. 3)[8]. Brynjolfsson, Rock, and Syverson (2021, p. 334)[9] explain why the payoff often arrives late or not at all: before AI can raise productivity, a firm must build assets that no balance sheet records, such as retrained staff and reorganized workflows. Lui, Lee, and Ngai (2022, p. 373)[10] find share prices dropping by 1.77% on average when firms announce AI investments. The drop is larger where the firm's IT capability is weak (Lui et al., 2022, p. 373)[11].

Recent theory explains why the results differ: the payoff depends on conditions inside the firm. A firm gains an advantage its rivals cannot copy when it *situates* AI in its own experience, structure, and relationships (Kemp, 2024, p. 618)[12]. Outcomes also depend on how a firm balances automation, where AI replaces human work, and augmentation, where AI supports it (Raisch & Krakowski, 2021, p. 192)[13]. Early empirical work supports this view: machine learning beat an older technology only when its users knew the domain well (Choudhury, Starr, & Agarwal, 2020, p. 1382)[14].

No systematic review has consolidated this evidence. A prominent review asks how

[6][Barney (1991), p. 110] "Several firms may all possess the same physical technology, but only one of these firms may possess the social relations, culture, traditions, etc. to fully exploit this technology in implementing strategies."

[7][Babina et al. (2024), p. 1] "AI-investing firms experience higher growth in sales, employment, and market valuations. This growth comes primarily through increased product innovation."

[8][Babina et al. (2024), p. 3] "the positive relationship between AI investments and firm growth is much stronger among ex-ante larger firms"

[9][Brynjolfsson, Rock & Syverson (2021), p. 334] "As firms adopt a new GPT, total factor productivity growth will initially be underestimated because capital and labor are used to accumulate unmeasured intangible capital stocks."

[10][Lui, Lee & Ngai (2022), p. 373] "The stock prices of the firms decrease by 1.77% on the day of the announcement."

[11][Lui, Lee & Ngai (2022), p. 373] "Nonmanufacturing firms and firms with weak information technology capabilities or low credit ratings suffer a more negative impact compared with other firms."

[12][Kemp (2024), p. 618] "The paper highlights the organizational activities involved in situating AI — specifically, grounding, bounding, and recasting."

[13][Raisch & Krakowski (2021), p. 192] "Overemphasizing either augmentation or automation fuels reinforcing cycles with negative organizational and societal outcomes."

[14][Choudhury, Starr & Agarwal (2020), p. 1382] "Domain expertise complements ML by correcting for the (strategic) incompleteness of the input to the ML tool, while vintage-specific skills ensure the ability to properly operate the technology."

firms adopt and use AI, not under which conditions AI leads to strategic outcomes (Enholm, Papagiannidis, Mikalef, & Krogstie, 2022, p. 1709)[15]. The same review concedes that "it is still not clear if and how AI can help organizations achieve financial performance gains" (Enholm et al., 2022, p. 1728)[16]. This thesis aims to close that gap. It collects and compares the firm-level evidence on the conditions under which AI investments turn into competitive advantage. These conditions include organizational capabilities, complementary assets, the design of the investment itself, and the firm's environment.

3 Research Question and Objectives

The thesis will answer the following research question:

> *Under what conditions do firm-level AI investments translate into competitive advantage and superior firm performance? — A systematic review of empirical evidence (2015–2026).*

Three filters narrow the question. First, the study must appear between 2015 and 2026. Second, it must study whole firms, not individuals or single tasks. Third, it must come from a top-ranked peer-reviewed journal (AJG 4* or 4, or VHB A+ or A). A paper from a lower-ranked journal is included only if it offers something the thesis needs, such as a unique data set. Every exception is documented with its reason. The thesis then has three goals: to map what the evidence says about how AI investment affects firm performance; to record the conditions that make this effect positive, negative, or absent; and to compare that evidence with current theory to find directions for future research.

4 Method

The thesis employs a systematic literature review (SLR) designed for maximal transparency and reproducibility (Tranfield, Denyer, & Smart, 2003, p. 209)[17]. The search will be conducted in Scopus (WU license); pilot searches via the Scopus API calibrated the following query:

[15][Enholm et al. (2022), p. 1709] "This study provides a systematic literature review that attempts to explain how organizations can leverage AI technologies in their operations and elucidate the value-generating mechanisms. Our analysis synthesizes the current literature and highlights: (1) the key enablers and inhibitors of AI adoption and use; (2) the typologies of AI use in the organizational setting; and (3) the first- and second-order effects of AI."

[16][Enholm et al. (2022), p. 1728] "Nevertheless, there is a long chain of causal associations, and to date it is still not clear if and how AI can help organizations achieve financial performance gains."

[17][Tranfield, Denyer & Smart (2003), p. 209] "Systematic reviews differ from traditional 'narrative' reviews by adopting a replicable, scientific and transparent process [...] that aims to minimize bias through exhaustive literature searches of published and unpublished studies and by providing an audit trail of the reviewers decisions, procedures and conclusions."

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TITLE(("artificial intelligence" OR "machine learning" OR "deep
learning" OR "AI")) AND TITLE-ABS-KEY(("invest" OR "invests" OR
"invested" OR "investing" OR "investment" OR "investments" OR
"investor" OR "investors" OR adopt* OR implement*) AND ("competitive
advantage" OR "firm performance" OR "financial performance" OR "firm
productivity" OR "firm value" OR "market value" OR "firm growth")) AND
SUBJAREA(BUSI OR ECON OR DECI) AND PUBYEAR > 2014 AND PUBYEAR < 2027
AND DOCTYPE(ar) AND LANGUAGE(english) AND SRCTYPE(j)
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As of 6 July 2026, the query returns 425 records, which lies inside the required corridor of 50–500 empirical studies. Before the search, three empirical studies were set as test cases that any sound query must find: Babina et al. (2024), Lui et al. (2022), and Krakowski, Luger, and Raisch (2023). The query finds all three, so it does not miss known relevant work. Included are empirical, firm-level, peer-reviewed journal articles in English published 2015–2026; excluded are conceptual papers, individual- or task-level studies, and non-ranked outlets. The screening process will be documented in a PRISMA-style flow diagram, so every decision can be reproduced.

Each included study is then coded into a state-of-the-art table: authors and year, journal and ranking, theoretical lens, method and sample, AI investment measure, performance outcome measure, direction of the effect, and identified conditions. This table is the core of the thesis.

5 Preliminary Structure of the Thesis

Chapter	**Content**
1. Introduction	Relevance of the topic, the research gap, and the research question.
2. Theoretical Background	AI as a general-purpose technology, resource-based reasoning, situated AI, and the automation–augmentation paradox.
3. Method	SLR protocol, search strategy, and coding scheme.
4. Results	Descriptive overview of the corpus and the state-of-the-art table.
5. Discussion	A framework of the conditions for AI-driven competitive advantage, compared with current theory.
6. Limitations	Constraints of the search, the corpus, and the reviewed studies.
7. Conclusion and Future Outlook	Summary of the findings and directions for future research.

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